

Notes: Gay, Gobbi & Goñi (JPE, 2026)

Revolutionary Transition: Inheritance Change and Fertility Decline

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	Inheritance-law atlas	4
3.2	Enquete Louis Henry database	4
3.3	Geni genealogies	4
3.4	Main outcomes and mechanisms	5
4	Models and empirical specifications	5
4.1	Difference-in-differences in the Henry data	5
4.2	Event-study specification	5
4.3	Flexible-trend DD	5
4.4	Spatial RD-DD in Geni	6
4.5	Heterogeneity by land-fragmentation costs	6
5	Identification and empirical design	6
5.1	Core identifying variation	6
5.2	Why difference-in-differences is plausible	6
5.3	Why RD-DD is useful	6
5.4	Mechanism identification	7
5.5	Main limitations	7
6	Section-by-section study guide	7
7	Tables and figures	8
7.1	Figure 1: Fertility decline in France, 1700-1850	8
7.2	Figure 2: Customs and inheritance systems in ancien regime France + Figure 3: Observations across egalitarian and inegalitarian inheritance	8
7.3	Figure 4: Completed fertility by cohort across inheritance systems + Figure 5: Fertility and distance to the inheritance border	9
7.4	Table 1: Balancedness of Prereform Characteristics + Table 2: DD Estimates, Henry Database	10
7.5	Table 3: Flexible-Trend DD Estimates, Henry Database + Table 4: Spatial RD Estimates, Geni Database	11
7.6	Table 5: Heterogeneous Effects by Soil Conditions for Small versus Large Farms	12
8	Result map	13

9 Short summary	13
10 Conclusion	14
10.1 Core conclusion	14
10.2 Economic intuition	14
10.3 Incremental contribution revisited	14
10.4 Why the paper matters	14
10.5 Limitations	14
10.6 Takeaway	14

1 Big picture

- **Question.** Did the French Revolution contribute causally to France’s early demographic transition by changing inheritance law?
- **Historical puzzle.** France experienced a major fertility decline well before industrialization, the conventional quantity-quality and human-capital mechanisms, and slow-moving cultural explanations can fully account for the timing.
- **Le Play hypothesis.** Le Play argued in 1875 that equal partition of inheritance forced households to limit fertility in order to avoid excessive fragmentation of family land.
- **Institutional shock.** The 1793 inheritance reforms imposed egalitarian inheritance across France: inheritance had to be partitioned equally among all children, including women.
- **Treatment variation.** Some regions already had egalitarian inheritance rules before the Revolution, while other regions had inegalitarian rules such as primogeniture, ultimogeniture, unigeniture, or female exclusion. The reforms therefore created a differential treatment across areas.
- **Main data innovation.** The authors construct a new local atlas of inheritance rules before the Revolution and combine it with individual-level fertility data from the Enquete Louis Henry and crowdsourced genealogies from Geni.
- **Main result.** Difference-in-differences estimates show that the reforms reduced completed fertility by about 0.5 children, increased childlessness, and delayed marriage.
- **External validation.** Regression-discontinuity difference-in-differences estimates near historical inheritance borders in the Geni data deliver similar, and in some specifications larger, fertility reductions.
- **Mechanism.** The strongest effects appear where soil and terrain favored small farms, consistent with households limiting fertility to prevent land fragmentation rather than mainly because women gained bargaining power.
- **Takeaway.** Legal institutions regulating inheritance can reshape fertility incentives and help trigger a demographic transition.

2 Incremental contribution of the paper

- **Direct empirical test of Le Play’s inheritance hypothesis.** The paper provides the first large-scale causal evidence that egalitarian inheritance reforms reduced fertility in France, converting a long-standing historical theory into a modern empirical test.
- **New institutional data infrastructure.** The authors construct a fine-grained map of 141 local customary and written laws regulating inheritance in ancien regime France. This is a major data contribution because earlier work usually distinguished only broad legal areas or regional customs.
- **Two independent fertility data sources.** The paper combines the nominative Enquete Louis Henry family-reconstitution data with large-scale genealogical records from Geni. The agreement between the two sources strengthens external validity.
- **Causal design linking law to demographic behavior.** The paper exploits differential exposure to the 1793 inheritance reforms across areas and cohorts, using both difference-in-differences and spatial RD-DD near inheritance borders.
- **Mechanism evidence on land fragmentation.** Heterogeneity by soil texture and terrain ruggedness links the fertility effect to the economic cost of splitting land across heirs, rather than only to broad cultural or bargaining-power interpretations.
- **Institutional explanation for the French demographic transition.** The paper explains both the timing and the abruptness of France’s fertility decline: a legal reform happened sharply around 1793 and changed household incentives quickly.

- **Contribution to development and family economics.** The findings show that inheritance institutions affect fertility, childlessness, and marriage timing, enriching theories of fertility transition beyond education, industrialization, and contraceptive technology.
- **Broader European relevance.** Since Napoleonic reforms spread egalitarian inheritance across Europe, the paper opens a channel through which institutional change may have contributed to the diffusion of demographic transition beyond France.

3 Data and measurement

3.1 Inheritance-law atlas

- The authors assemble original documents codifying local inheritance customs and laws before the French Revolution.
- They classify inheritance rules along two dimensions: partible versus impartible inheritance and whether women were included or excluded from inheritance.
- Regions treated by the 1793 reform are those where prerevolutionary rules were unequalitarian: impartible inheritance and/or female exclusion.
- Regions untreated by the reform already complied with egalitarian inheritance before 1793.

Interpretation: This map is essential because treatment is local and institutional, not simply regional. It distinguishes areas where the Revolution changed inheritance incentives from areas where it mainly codified existing practices.

3.2 Enquete Louis Henry database

- The Henry data use family reconstitution from parish, hospital, and civil records.
- The baseline sample includes women born from 1700 to 1803 in 39 rural municipalities.
- Main outcomes include completed fertility, completed fertility of mothers, childlessness, and age at marriage.
- The baseline estimation uses cohorts with different exposure to the 1793 reform: cohorts completing fertility before the reform versus cohorts still fertile after the reform.

Interpretation: The Henry data are high quality and allow individual-level controls and municipality fixed effects, but the sample is geographically limited.

3.3 Geni genealogies

- The Geni data are crowdsourced genealogical records covering a much larger geography.
- The RD-DD analysis uses mothers born in France between 1700 and 1810 within a bandwidth around historical inheritance borders.
- The horizontal restriction uses border proximity to compare places close to different inheritance regimes.
- The timing restriction compares cohorts fertile before and after the 1793 reform.

Interpretation: The Geni data trade some precision for spatial breadth. They make it possible to run a border design around many inheritance-regime boundaries.

3.4 Main outcomes and mechanisms

- **Completed fertility:** total births over a woman’s lifetime, excluding children who died before age six in the net-fertility specification.
- **Completed fertility of mothers:** fertility conditional on having at least one child.
- **Childlessness:** indicator for no surviving children.
- **Age at marriage:** mechanism outcome capturing fertility delay.
- **Soil and ruggedness:** proxies for whether land fragmentation is especially costly for small farms.

Interpretation: The outcomes distinguish intensive-margin fertility decline, extensive-margin childlessness, marriage timing, and the land-fragmentation mechanism.

4 Models and empirical specifications

4.1 Difference-in-differences in the Henry data

The baseline specification compares treated and untreated inheritance areas across cohorts fertile before and after 1793:

$$Y_{imc} = \alpha_m + \delta_c + \beta(Reform_m \times FertilePost_c) + X'_{imc}\Gamma + \varepsilon_{imc}.$$

- Y_{imc} is fertility or marriage outcome for woman i in municipality m and cohort c .
- $Reform_m$ equals one if the municipality had inequalitarian inheritance before 1793.
- $FertilePost_c$ captures cohorts still fertile after the 1793 inheritance reforms.
- Municipality fixed effects absorb time-invariant differences between inheritance areas.
- Cohort fixed effects absorb aggregate fertility trends.

Interpretation: The coefficient β asks whether fertility fell more among cohorts exposed to reform in treated areas than in already-egalitarian areas.

4.2 Event-study specification

The event-study version interacts treatment with cohort exposure bins:

$$Y_{imc} = \alpha_m + \delta_c + \sum_k \beta_k(Reform_m \times CohortBin_{k,c}) + X'_{imc}\Gamma + \varepsilon_{imc}.$$

Interpretation: The event study tests whether pretrends are flat before the reform and whether effects emerge for cohorts with fertile years after 1793.

4.3 Flexible-trend DD

The flexible-trend specification allows prereform covariates to have cohort-specific effects:

$$Y_{imc} = \alpha_m + \delta_c + \beta(Reform_m \times FertilePost_c) + Z_m \times \delta_c + X'_{imc}\Gamma + \varepsilon_{imc}.$$

Interpretation: This controls for differential fertility trends by wheat prices, population density, religiosity, distances to administrative and information centers, political exposure, and other municipal characteristics.

4.4 Spatial RD-DD in Geni

Near inheritance borders, the paper estimates spatial RD and RD-DD models:

$$Y_{ijc} = \alpha_b + \delta_c + \beta Reform_j + f(Distance_j) + \varepsilon_{ijc}$$

for prereform or postreform cohorts, and

$$Y_{ijc} = \alpha_b + \delta_c + \beta_1(Reform_j \times FertilePost_c) + f(Distance_j) + \varepsilon_{ijc}$$

for the RD-DD design.

Interpretation: RD-DD compares fertility changes over time for communities located close to opposite sides of an inheritance border, reducing concerns about broad regional differences.

4.5 Heterogeneity by land-fragmentation costs

$$Y_{imc} = \alpha_m + \delta_c + \beta_1(Reform_m \times FertilePost_c \times SmallFarmConditions_m) + \beta_2(Reform_m \times FertilePost_c \times LargeFarmConditions_m) + \varepsilon_{imc}$$

Interpretation: Soil texture and ruggedness proxy for whether dividing land among many heirs is especially costly. Stronger effects in small-farm conditions support the land-fragmentation mechanism.

5 Identification and empirical design

5.1 Core identifying variation

The main identifying variation is the interaction of geography and cohort exposure. Geography determines whether the 1793 reform changed the local inheritance regime. Cohort timing determines whether a woman was still fertile when the reform occurred.

5.2 Why difference-in-differences is plausible

- Treated and untreated areas are compared before and after the same national reform.
- Municipality fixed effects absorb time-invariant differences between inheritance regions.
- Cohort fixed effects absorb aggregate fertility decline and national shocks.
- Flexible trends address differential changes by prereform covariates.
- The event-study figure shows whether effects predate the reform.

5.3 Why RD-DD is useful

- Border comparisons focus on places geographically close to each other but governed by different inheritance customs.
- The RD component tests whether prereform fertility differs across inheritance borders.
- The RD-DD component tests whether the border discontinuity changes after the reform.
- This helps separate inheritance-rule effects from broad regional culture and geography.

5.4 Mechanism identification

The paper cannot randomly assign land fragmentation costs, but it uses soil and terrain to test where inheritance reform should matter most. If households reduced fertility to avoid land fragmentation, effects should be stronger where land was more naturally suited to small farms and weaker where large farms were favored. Table 5 supports this prediction.

5.5 Main limitations

- The Henry sample is limited to 39 rural municipalities.
- Geni genealogies are large but may have coverage and accuracy issues.
- Treatment combines multiple legal changes: abolition of impartibility and inclusion of women.
- The paper argues against a bargaining-power mechanism, but it cannot directly observe household bargaining or land-transfer decisions for every family.

6 Section-by-section study guide

- **Introduction.** Read for the puzzle: France’s fertility decline is early and sharp, and inheritance law offers a timing-consistent explanation.
- **Historical background.** Focus on prerevolutionary inheritance variation and the 1793 reforms that imposed equal partition including women.
- **Data.** Understand the two fertility sources: Henry for high-quality individual reconstruction and Geni for broad spatial coverage.
- **DD results.** Tables 1-3 and Figure 4 show balance, baseline DD, flexible controls, and event-study dynamics.
- **RD-DD results.** Figure 5 and Table 4 provide spatial border evidence using Geni.
- **Mechanism.** Table 5 is crucial for land-fragmentation evidence; the discussion of women’s bargaining power rules out an alternative channel.
- **Conclusion.** Note the headline claim: the reform accounts for a large share of the aggregate fertility decline and puts legal institutions at the center of the French demographic transition.

7 Tables and figures

7.1 Figure 1: Fertility decline in France, 1700-1850

Read:

- The figure plots the crude birth rate, Princeton marital fertility index, and completed fertility of mothers aged 40.
- A vertical line marks the 1793 inheritance reforms.
- Fertility trends down slowly before the Revolution and then falls sharply after the reforms.
- The visual timing motivates the paper’s legal-institution explanation.

Detailed interpretation: The figure is not a causal estimate; it is a historical motivation. Its importance is timing: the fertility drop begins too early for industrialization and too abruptly for slow cultural change. The sharp post-1793 break suggests a national institutional shock could matter.

Economic message: The demographic transition may have institutional roots: legal rules changed fertility incentives before industrialization.

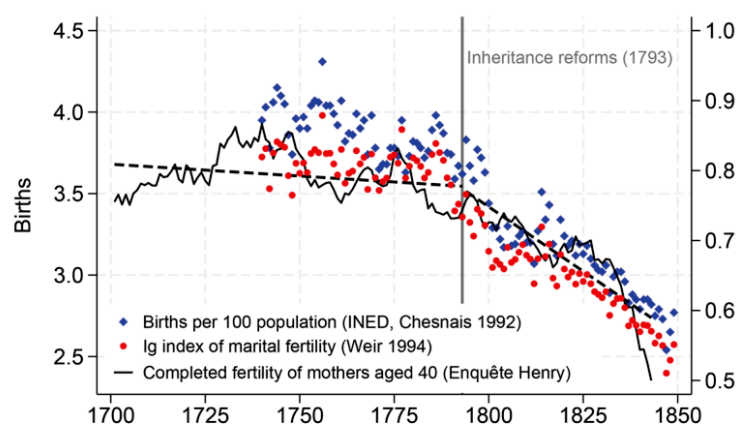


FIG. 1.—Fertility decline in France, 1700–1850. I_x is the Princeton index of fertility, that is, the number of legitimate births relative to the expected number of a Hutterite population with the same number and age structure of married women. Completed fertility is a 5-year moving average plotted in the year when mothers were 40. Dashed lines are fitted values relative to completed fertility before and after 1793.

Figure 1: Fertility decline in France, 1700-1850.

7.2 Figure 2: Customs and inheritance systems in ancien regime France + Figure 3: Observations across egalitarian and inegalitarian inheritance

Read:

- Figure 2 maps customary versus written law areas and inheritance systems by partibility and women’s inclusion.
- Yellow lines mark areas treated by the 1793 inheritance reform.
- Figure 3 maps observations in the Henry DD sample and the Geni RD-DD sample.
- Together, the figures show both treatment assignment and empirical coverage.

Detailed interpretation: Figure 2 is the institutional backbone of the paper. It turns legal history into a spatial treatment variable. Figure 3 then shows that the data cover both treated and untreated areas. The Henry sample is smaller but carefully reconstructed; the Geni sample is much larger and supports border-based validation.

Economic message: The paper’s contribution relies on building new historical-geographic data: without the inheritance map, the treatment would be invisible.

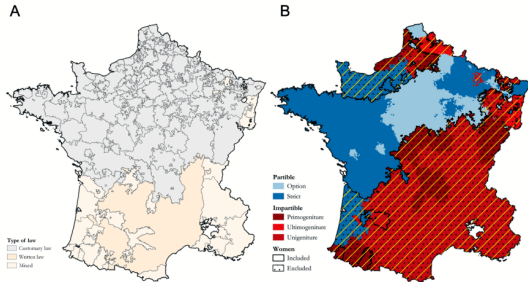


FIG. 2.—Customs and inheritance systems in ancien régime France. A, Spatial distribution of customary law and written law areas in ancien régime France. B, Spatial distribution of inheritance systems across partibility and women inclusion. Areas treated by the 1793 inheritance reforms are highlighted in yellow.

(a) Figure 2: customs and inheritance systems.

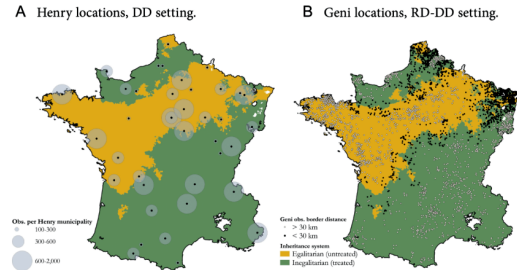


FIG. 3.—Observations across egalitarian and inegalitarian inheritance. A, Distribution of observations in the Henry database used in the DD setting. B, Distribution of observations in the Geni database used in the RD-DD setting. Yellow areas denote egalitarian inheritance systems before the Revolution, and green areas denote inegalitarian inheritance systems.

(b) Figure 3: Henry and Geni observation locations.

7.3 Figure 4: Completed fertility by cohort across inheritance systems + Figure 5: Fertility and distance to the inheritance border

Read:

- Figure 4 shows fertility trends by cohort for partible/female-inclusive versus impartible/female-exclusive inheritance systems.
- Event-study estimates show little evidence of prereform treatment effects and negative effects for exposed cohorts.
- Figure 5 shows border RD plots before and after 1793.
- The prereform discontinuity is large, but after 1793 the gap closes, consistent with convergence after reform.

Detailed interpretation: Figure 4 is the DD visual: treated and untreated areas have different fertility levels before the reform, but the key evidence is the post-1793 closure of the gap among fertile cohorts. Figure 5 is the spatial validation: near inheritance borders, fertility differences shrink after the reform, reducing concerns that broad geography drives the result.

Economic message: The reforms caused convergence in fertility across previously different inheritance regimes.

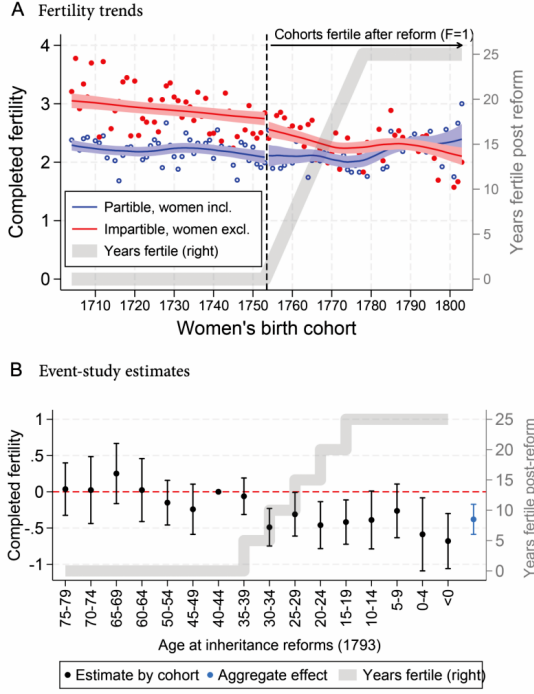
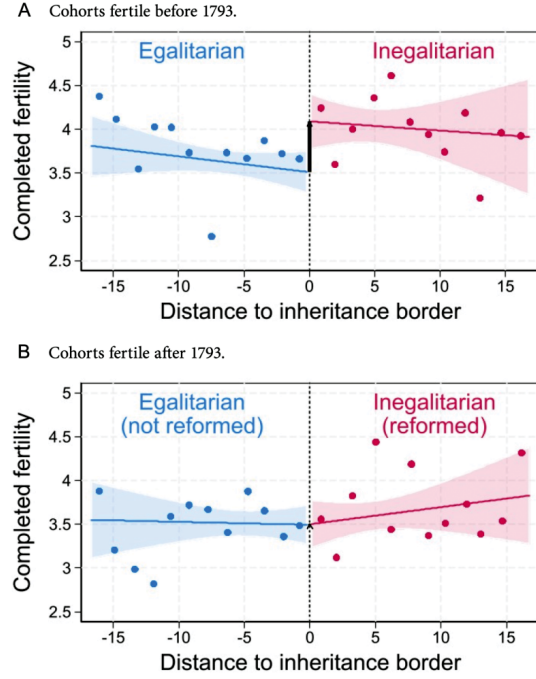


FIG. 4.—Completed fertility by cohort across inheritance systems. A. Average completed fertility (points), trends (lines), and 95% confidence intervals (shaded areas) from a local polynomial regression on each side of the 1753 birth cohort. B. β coefficients and 95% confidence intervals for equation (9) as a sample of all women aged 0–70 at the time of the

(a) Figure 4: cohort fertility and event study.



(b) Figure 5: spatial RD fertility at inheritance borders.

7.4 Table 1: Balancedness of Prereform Characteristics + Table 2: DD Estimates, Henry Database

Read:

- Table 1 shows prereform differences between egalitarian and inegalitarian inheritance areas.
- Completed fertility is higher in inegalitarian areas before reform, while most other demographic and municipal characteristics are relatively balanced.
- Table 2 reports DD estimates: completed fertility falls by about 0.48-0.52 children in treated areas for exposed cohorts.
- The reform also lowers fertility among mothers by about 0.35-0.39 children and raises childlessness by about 6.2-7.6 percentage points.

Detailed interpretation: Table 1 explains why a simple cross-sectional comparison would be misleading: inegalitarian areas have higher prereform fertility. Table 2 uses within-municipality and cohort variation to show that the reform reduced fertility precisely where inheritance law changed. The childlessness result shows both extensive-margin and intensive-margin fertility responses.

Economic message: Egalitarian inheritance reform lowered fertility substantially, not just by changing composition but by changing fertility behavior among exposed cohorts.

BALANCEDNESS OF PREREFORM CHARACTERISTICS						
	Prereform Inheritance				Difference	
	Egalitarian		Inegalitarian			
	Mean	SD	Mean	SD	Mean	SE
A. Prereform Individual-Level Characteristics						
Completed fertility	2.22	2.28	2.84	2.72	.62	.16***
Wife's age at death	56.96	17.08	58.51	17.57	1.55	1.56
Husband's age at death	59.72	15.53	61.59	15.80	1.87	1.27
Wife died before age 40	.14	.34	.12	.32	-.02	.02
Wife's mother alive at marriage	.51	.50	.47	.50	-.04	.06
Husband's mother alive at marriage	.45	.50	.41	.49	-.04	.05
Wife's father alive at marriage	.42	.49	.41	.49	-.01	.05
Husband's father alive at marriage	.38	.48	.36	.48	-.01	.04
Known birth year or in general population census	.63	.48	.63	.48	-.00	.05
Known union end date	.73	.44	.67	.47	-.06	.03
Wife's literacy	.16	.36	.09	.29	-.06	.06
Husband's literacy	.37	.48	.29	.45	-.08	.10
Observations	5,870		4,882		10,752	
B. Prereform Municipality-Level Characteristics						
Wheat price (log)	1.98	.09	2.07	.23	.09	.06
Population density (log)	3.97	.54	3.87	.59	-.10	.18
Religiosity index	.56	.30	.53	.28	-.03	.09
Distance to religious center	3.18	.61	3.15	.83	-.04	.23
Distance to judicial district seat	2.37	.78	2.66	.65	.29	.23
Distance to territorial administration	2.55	.69	2.47	.55	-.07	.20
Distance to tax center	2.69	.74	2.83	.70	.14	.23
Distance to political society	1.74	.64	1.67	.66	-.07	.21
Distance to rebellion in 1780–89	2.97	1.05	2.71	.77	-.26	.29
Distance to paved road	.81	.62	.90	.67	.09	.21
Distance to horse post relay	2.16	.60	2.63	.68	.46	.21**
Average caloric suitability of land	1,902	305	1,868	396	-.34	114
Terrain ruggedness (in 100s of meters)	.51	.31	.76	.52	.25	.15
Share sandy soils	.17	.32	.13	.25	-.04	.09
Observations	20		19		39	

NOTE.—In panel A, the sample is the 10,752 women in the Henry database who completed their reproductive span before the 1793 inheritance reforms. In panel B, the sample is the 39 municipalities in the Henry database. Age at death is missing for about 30% of wives and 40% of husbands. Known birth year is recovered from parish records or from age in the general population census. Distances are in log(1 + km) units. For terrain ruggedness, the sample excludes three villages in the Alps and Pyrenees with extreme values (twice as large as the 90th percentile value). Standard errors are clustered by municipality in panel A and are robust standard errors in panel B.

** $p < .05$.

(a) Table 1: balancedness of prereform characteristics.

DD ESTIMATES, HENRY DATABASE				
	(1)	(2)	(3)	(4)
A. Dependent Variable: Completed Fertility				
Reformed inheritance × fertile postreform	-.492*** (.117)	-.493*** (.118)	-.475*** (.113)	-.520*** (.127)
Observations	20,332	20,332	20,331	20,322
Adjusted R^2	.044	.044	.152	.175
B. Dependent Variable: Completed Fertility of Mothers				
Reformed inheritance × fertile postreform	-.354** (.142)	-.356** (.148)	-.371** (.151)	-.392** (.160)
Observations	14,066	14,066	14,065	14,057
Adjusted R^2	.057	.058	.083	.091
C. Dependent Variable: =1 If Childless				
Reformed inheritance × fertile postreform	.067*** (.017)	.067*** (.017)	.062*** (.013)	.076*** (.017)
Observations	20,332	20,332	20,331	20,322
Adjusted R^2	.023	.024	.160	.186
Cohort fixed effects	Yes	Yes	Yes	Yes
Municipality fixed effects	Yes	Yes	Yes	Yes
Literacy (0/1)		Yes	Yes	Yes
Accuracy of Henry form fixed effects			Yes	Yes
Father alive at marriage (0/1)				Yes
Mother alive at marriage				Yes
N clusters	39	39	39	39

NOTE.—The sample is women born in 1700–1803 in the Henry database. The dependent variable is the number of children ever born to all women (panel A), to mothers (panel B), and the probability to be childless (panel C). All variables consider “net” fertility, i.e., they are based on the number of children who reached age 6. Literacy is an indicator variable equal to 1 if the woman signed the marriage certificate. Accuracy of Henry form includes fixed effects for each category listed in table A2. Standard errors in parentheses are clustered by municipality.

** $p < .05$.

*** $p < .01$.

Panels B and C of table 2 report estimates for the effect of the 1793 in-

(b) Table 2: DD estimates, Henry database.

7.5 Table 3: Flexible-Trend DD Estimates, Henry Database + Table 4: Spatial RD Estimates, Geni Database

Read:

- Table 3 adds flexible cohort-specific trends by prereform covariates.
- The fertility effect remains about -0.43 to -0.50 children, and age at marriage rises by about 1.94 years.
- Table 4 uses Geni data near inheritance borders.
- The RD-DD estimates are negative and significant, around -0.85 to -1.20 children depending on controls and bandwidths.

Detailed interpretation: Table 3 addresses the concern that treated areas were already on different trends because of geography, religiosity, prices, administrative proximity, or political exposure. Table 4 addresses a different concern: regional comparability. The RD-DD design compares close-by places on opposite sides of inheritance borders, making the same qualitative conclusion in a much larger genealogy sample.

Economic message: The result survives two distinct identification strategies: cohort-by-region DD and spatial border RD-DD.

TABLE 3
FLEXIBLE-TREND DD ESTIMATES, HENRY DATABASE

	Completed Fertility			Completed Fertility of Mothers	=1 If Childless	Age at Marriage
	(1)	(2)	(3)	(4)	(5)	(6)
Reformed inheritance × fertile postreform	−.478*** (.174)	−.504*** (.159)	−.434*** (.154)	−.432*** (.201)	.046** (.021)	1.938*** (.572)
Observations	20,322	20,322	20,322	14,057	20,322	20,321
N clusters	39	39	39	39	39	39
Adjusted R^2	.178	.179	.182	.103	.190	.255
Cohort and municipality fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Individual-level controls	Yes	Yes	Yes	Yes	Yes	Yes
Wheat price series	Yes	Yes	Yes	Yes	Yes	Yes
Cohort fixed effects × wheat price 1780s	Yes	Yes	Yes	Yes	Yes	Yes
Cohort fixed effects × population density 1793	Yes	Yes	Yes	Yes	Yes	Yes
Cohort fixed effects × religiosity index		Yes	Yes	Yes	Yes	Yes
Cohort fixed effects × distance religious center		Yes	Yes	Yes	Yes	Yes
Cohort fixed effects × distance political society			Yes	Yes	Yes	Yes
Cohort fixed effects × distance rebellion			Yes	Yes	Yes	Yes
Cohort fixed effects × economic geography variables			Yes	Yes	Yes	Yes
Cohort fixed effects × network variables			Yes	Yes	Yes	Yes

NOTE.—The sample is women born in 1700–1803 in the Henry database. The dependent variable is the number of children ever born to all women in cols. 1–3, to mothers in col. 4, the probability to be childless in col. 5, and age at marriage in col. 6. All variables consider “net” fertility, i.e., they are based on the number of children who reached age 6. All specifications include cohort fixed effects and the full set of individual-level controls in table 2. The remaining covariates capture flexible trends in fertility by municipality-level characteristics. Economic geography variables include the distance to legal centers, tax

(a) Table 3: flexible-trend DD estimates.

TABLE 4
SPATIAL RD ESTIMATES, GENI DATABASE

	RD Prereform	RD Postreform	RD-DD All			
	(1)	(2)	(3)	(4)	(5)	(6)
Estimate	1.044*** (.368)	.186 (.354)	−.848** (.365)	−.928** (.382)	−1.187** (.557)	−1.197*** (.445)
Observations	2,385	1,848	4,325	3,707	3,527	5,410
N clusters	385	624	959	862	778	1,214
Mean dependent variable	3.81	3.58	3.69	3.77	3.85	3.63
Cohort fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Border segment fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Judicial district fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Flexible trends					Yes	Yes
Order polynomial	Linear	Linear	Linear	Linear	Linear	Quadratic
MSR bandwidth (km)	19.47	19.47	19.47	16.91	17.49	29.29

NOTE.—This table reports estimates of β and δ in eqs. (4) and (5). The sample is mothers whose Geni record satisfies the horizontal restriction, born in France (1700–1810) within a MSR-optimal bandwidth on each side of the inheritance border. We use local polynomial fits of orders 1 and 2 and triangular kernel functions for local polynomial estimation. The dependent variable is the number of children ever born to mothers, excluding child deaths before age 6. Flexible trends include municipality-level wheat prices by decade and cohort fixed effects interacted with a vector of initial municipality-level variables: wheat price in 1780, population density in 1793, the share of refractory clergy in 1791, and indicator variables I_k within 15 kilometers, there was a political society; rebellions against state authorities in the 1780s; religious, legal, fiscal, or territorial administrative centers; horse post relay; or paved roads (within 7.5 kilometers). Standard errors in parentheses are clustered by municipality.

** $p < .05$.

*** $p < .01$.

(b) Table 4: spatial RD estimates, Geni database.

7.6 Table 5: Heterogeneous Effects by Soil Conditions for Small versus Large Farms

Read:

- Table 5 splits the reform effect by whether soil and terrain favored small or large farms.
- In places where soil conditions favor large farms, reform effects are close to zero and statistically insignificant.
- In places where soil conditions favor small farms, effects are about -0.46 to -0.61 children and statistically significant.
- The differential effect is significant in several specifications.

Detailed interpretation: This is the mechanism table. If inheritance reform reduced fertility because households feared land fragmentation, effects should be stronger where splitting land into smaller parcels is especially costly. That is what the table shows. If the mechanism were mainly women’s bargaining power, the soil-based heterogeneity would be harder to explain.

Economic message: The paper’s strongest mechanism evidence supports land-fragmentation incentives rather than broad cultural change or female empowerment.

TABLE 5
HETEROGENEOUS EFFECTS BY SOIL CONDITIONS FOR SMALL VERSUS LARGE FARMS

	Texture		Ruggedness	
	(1)	(2)	(3)	(4)
Reform $\times$ fertile (soil conditions for large farms)	-.015 (.220)	-.093 (.262)	-.133 (.218)	.063 (.216)
Reform $\times$ fertile (soil conditions for small farms)	-.505*** (.144)	-.456*** (.150)	-.575*** (.204)	-.614*** (.194)
Differential effect	.490	.363	.442	.677
<i>p</i> -value difference	.031	.218	.097	.008
Cohort and municipality fixed effects	Yes	Yes	Yes	Yes
Individual-level controls	Yes	Yes	Yes	Yes
Flexible trends	Yes	Yes	Yes	Yes
Texture $\times$ cohort fixed effects		Yes		Yes
Ruggedness $\times$ cohort fixed effects		Yes		Yes
Observations	20,322	20,322	20,322	20,322
Clusters	39	39	39	39

NOTE.—The dependent variable is completed fertility. In cols. 1 and 2, soil conditions are based on texture, where sandy soils favor large farms and nonsandy soils favor small farms. In cols. 3 and 4, soil conditions are based on ruggedness, where flat terrains favor large farms and rugged terrains favor small farms. The sample is women born in 1700–1803 in the Henry database. Individual-level controls and flexible trends are defined as in table 3. Columns 2 and 4 account for lower-order interactions by including soil sandiness and terrain ruggedness interacted with cohort fixed effects. Constants are omitted. Standard errors in parentheses are clustered by municipality.
*** $p < .01$.

Table 5: heterogeneous effects by soil conditions.

8 Result map

- **Historical pattern:** French fertility falls sharply soon after the French Revolution, before industrialization.
- **Treatment construction:** prerevolutionary inheritance rules determine whether the 1793 reforms changed local inheritance incentives.
- **Baseline Henry DD:** reform exposure reduces completed fertility by roughly half a child.
- **Extensive margin:** childlessness increases, indicating deliberate fertility limitation and not only fewer births among mothers.
- **Timing:** event-study estimates show effects for cohorts fertile after the reform rather than smooth prereform divergence.
- **Border validation:** Geni RD-DD estimates show the same fertility convergence near inheritance borders.
- **Mechanism:** effects are concentrated where land fragmentation was most costly, supporting the Le Play land-inheritance channel.
- **Alternative channel:** qualitative and quantitative evidence suggests women’s bargaining power did not materially increase after the reform.
- **Aggregate importance:** the authors argue that the reforms account for a large share of France’s aggregate fertility decline.

9 Short summary

- The paper tests whether inheritance reform contributed to France’s early demographic transition.
- The 1793 French revolutionary reforms imposed egalitarian inheritance across children and women.
- The authors construct new local inheritance-rule data for ancien regime France.
- They combine this map with the Henry family-reconstitution data and Geni genealogies.

- Difference-in-differences estimates show that exposed cohorts in treated areas reduced fertility by about 0.5 children.
- Event studies support timing around the reform.
- RD-DD estimates near inheritance borders support the same conclusion.
- Effects are stronger where soil and terrain favored small farms, consistent with land-fragmentation incentives.
- The evidence does not strongly support a women’s-bargaining-power mechanism.
- The paper places legal institutions at the center of France’s early fertility transition.

10 Conclusion

10.1 Core conclusion

Gay, Gobbi, and Goñi show that the French Revolution’s inheritance reforms were a major cause of France’s early fertility decline. By requiring equal inheritance among children, including women, the reform increased the cost of having many heirs in places that previously used inegalitarian inheritance rules.

10.2 Economic intuition

The core intuition is intergenerational land fragmentation. If land must be split equally among many children, parents with more children risk dividing family wealth into plots too small to sustain descendants. Reducing fertility becomes a way to preserve family assets across generations.

10.3 Incremental contribution revisited

The paper’s incremental contribution is to provide causal empirical evidence for a historically important but previously untested hypothesis: inheritance institutions can drive fertility transitions. It does so with new legal-geographic data, two independent demographic sources, two identification strategies, and a mechanism test based on land-fragmentation costs.

10.4 Why the paper matters

The paper changes how we think about demographic transition. It shows that fertility decline can come from legal institutions that alter family incentives, not only from industrialization, education, mortality decline, contraception, or culture. It also shows that abrupt institutional changes can produce rapid demographic responses.

10.5 Limitations

The Henry data cover only a set of rural municipalities, while the Geni data are broader but crowdsourced. The treatment combines multiple inheritance changes, and the authors cannot observe every household’s landholding strategy. The mechanism evidence is compelling but indirect: land-fragmentation incentives are inferred from soil and terrain conditions.

10.6 Takeaway

Legal institutions can reshape fertility decisions. In late eighteenth-century France, egalitarian inheritance law reduced fertility by changing the economic consequences of family size.